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Implementation of a Frama-C plugin and a language server for ACSL/C code assistance

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Internship report

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1 Introduction

1.1 Thales DIS, IBS

Thales Digital Identity and Security is the global business unit (GBU) of Thales which operates in the field of secure software, providing biometrics and encryption solutions. Within the DIS GBU is the Identity and Biometric Solutions business line (BL) which is in charge of digital identity in a government scale and operate on cards and passports. the IBS BL is divided in three subunits : ID Documents Solutions (IDDS), Digital ID & Verification Solutions and Public Security. I am assigned to the IDDS subunit as an intern software developer, under the supervision of Adel Djoudi, formal methods expert at Thales.

1.2 Background

In the IDDS subunit, developers work using program analyzers such as Frama-C, a C code analyzer developed by CEA and Inria. This software provides a precise analysis of their code and potential rifts in it by manipulating functional specification in ACSL language. But Frama-C currently does not provide code comprehension and coding features for C and ACSL, some VS Code extensions exist but they are not able to fully satisfy the needs of developers (why??). The goal of this project is to provide a code assistance tool (reecrire sujet dans la convention de stage)

2 Integration into the company

During my first day at Thales, I was welcomed at the reception and they instantly made me a [badge] with authorizations specific to my role, there are places in can access and others that i can't. M. Djoudi accompanied me through the whole process and introduced me to the IDDS team. Thales DIS gives a lot of attention to any security threats that could occur physically or virtually.

3 Documentation

3.1 The Language Server Protocol

Developers working in large scale projects often need a code editing environment with the adequate tooling for the specific languages they use in their workspace. Multiple solutions such as Eclipse or IntelliJ IDEA exist for Java projects, as well as other popular programming languages. But the complexity of developing one tool for one language and one specific editor is unreasonable as it forces developers to repeat the same document analysis logic for every code editor or Integrated Development Environement (IDE) with each programming language.

3.2 Frama-C plugin development

- how is frama c structured - what is a plugin and why do we need one in our case

[1] [2] [3]

References

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- [2] Julien Signoles et al. *Frama-C Plug-in Development Guide*. URL: <http://frama-c.com/download/frama-c-plugin-development-guide.pdf>.
- [3] George C. Necula et al. "CIL: Intermediate Language and Tools for Analysis and Transformation of C Programs". In: *Compiler Construction*. Ed. by R. Nigel Horspool. Berlin, Heidelberg: Springer Berlin Heidelberg, 2002, pp. 213–228. ISBN: 978-3-540-45937-8.